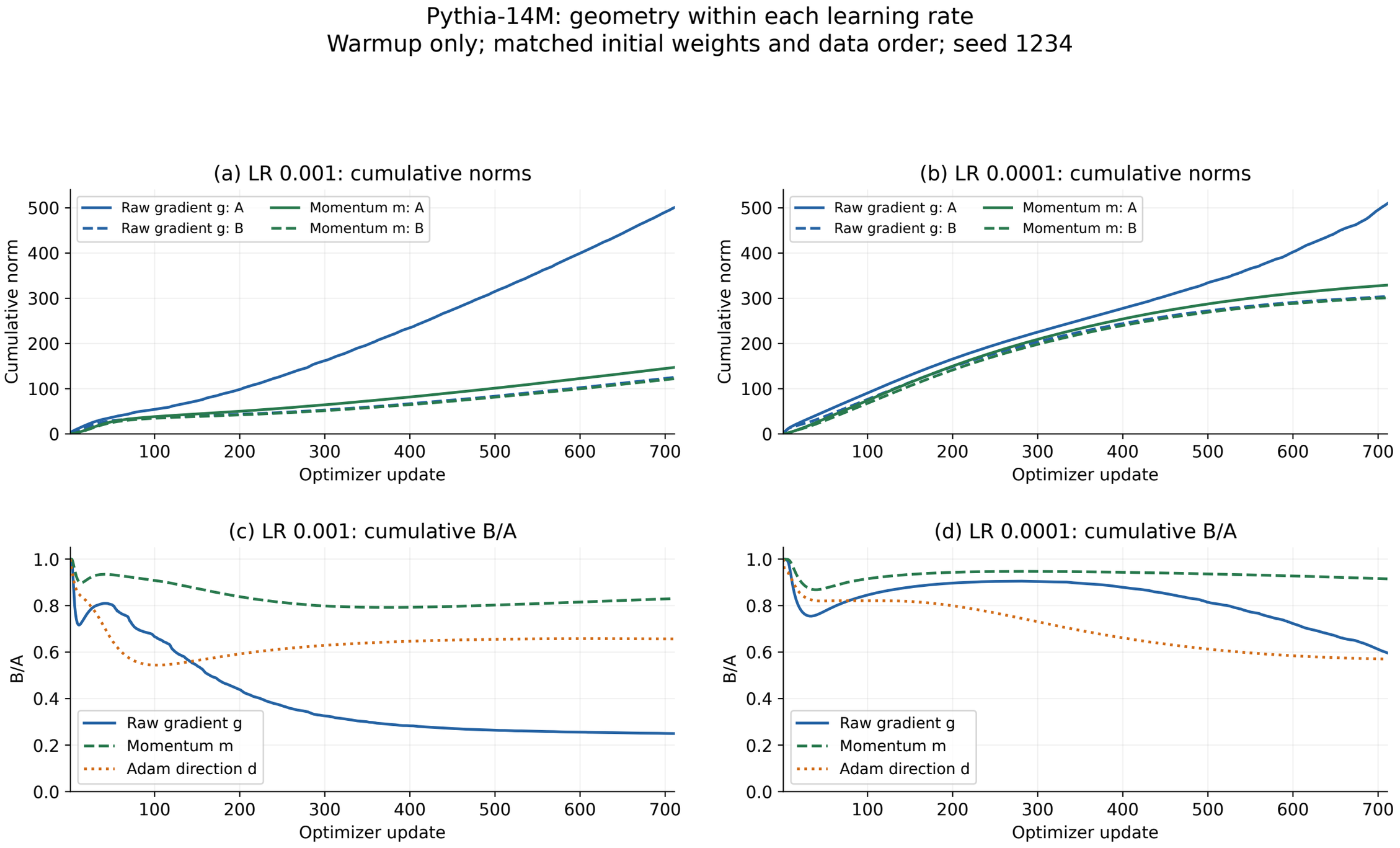
Pythia-14M: geometry within each learning rate
Warmup only; matched initial weights and data order; seed 1234
(a) LR 0.001: cumulative norms
(b) LR 0.0001: cumulative norms
Raw gradient g: A
Raw gradient g: B
Momentum m: A
Momentum m: B
Cumulative norm
Optimizer update
(c) LR 0.001: cumulative B/A
(d) LR 0.0001: cumulative B/A
Raw gradient g
Momentum m
Adam direction d
B/A
A = sum of vector norms; B = norm of vector sum. Colors identify raw gradient g, stored momentum m and adaptive direction d.
Cumulative norms: solid A, dashed B. Both rows share scales across learning rates. Display: full-sized updates 1-711.
All curves are unsmoothed. Partial update 712 remains in numerical results. One matched seed; no cross-seed uncertainty estimated.